

Quantitative Well-Posedness for Mean-Field Games with Common Noise: An Explicit Contraction Estimate

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Abstract

We study the well-posedness of mean-field games (MFGs) with common noise, in which all agents are subject to a shared aggregate shock in addition to idiosyncratic noise. While existence and uniqueness for such systems is known under linear-convex and weak-monotonicity structural conditions, the dependence of the solution’s stability on the model’s primitive constants – the Lipschitz constant of the drift, the magnitude of the common-noise coefficient, and the regularity of the Lions derivatives – has not been made explicit. We prove that the mean-field fixed-point map is a contraction in a weighted Wasserstein-2 metric with an explicit constant $C = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}$, obtained via a synchronous-coupling argument that isolates the first-order cancellation of the common noise in expectation from its residual second-order contribution. The explicit constant directly informs the choice of discretisation step and learning rate in downstream numerical and reinforcement-learning algorithms for this class of games. We verify the ε -Nash property with rate $O(N^{-1/2})$ and discuss the scope and limitations of the result, including the gap between the expectation-level contraction established here and pathwise stability.

Keywords: mean-field games, common noise, McKean–Vlasov FBSDEs, Wasserstein contraction, ε -Nash equilibrium

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1 Introduction

Mean-field games (MFGs), introduced independently by Lasry and Lions (2007) and Huang et al. (2006), model the limiting behaviour of symmetric N -player stochastic differential games as $N \rightarrow \infty$. In the canonical formulation, each agent’s idiosyncratic noise is averaged away by the law of large numbers, and the population measure flow $\mu = (\mu_t)_{t \in [0, T]}$ becomes deterministic. *Common noise* – an aggregate shock affecting all agents simultaneously, such as a macroeconomic or market-wide disturbance – breaks this determinism: the population measure μ_t becomes itself a stochastic process, adapted to the common-noise filtration. This single structural change is consequential: it transforms the deterministic Fokker–Planck PDE governing μ_t into a stochastic PDE, and the mean-field fixed-point argument that establishes existence and uniqueness must be revisited at the level of conditional laws rather than deterministic measures.

Carmona and Delarue (2018) provide the comprehensive probabilistic treatment of MFGs with common noise via the master equation and the associated conditional McKean–Vlasov FBSDE

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system. Ahuja et al. (2019) establish existence and uniqueness for a class of common-noise MFGs with linear-convex structure and weak-monotonicity, via a continuation method applied to the FBSDE system. These results establish *qualitative* well-posedness: a solution exists and is unique. What remains open – and is the focus of the present paper – is the *quantitative* dependence of the solution’s stability on the primitive model constants.

Contribution. We prove (Theorem 4.1) that the mean-field fixed-point map Φ is a contraction in a weighted Wasserstein-2 norm with an explicit constant

$$C = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}},$$

where L_b is the Lipschitz constant of the drift in (x, μ, α) , $\|\sigma_0\|_F$ is the Frobenius norm of the common-noise diffusion coefficient, and C_{reg} collects the regularity constants of the Lions derivatives of the model coefficients and the associated BSDE estimates. The proof uses a *synchronous coupling* argument: by driving the two candidate measure flows’ representative-agent systems with the *same* realisations of both the idiosyncratic noise W^i and the common noise W^0 , the first-order stochastic term arising from σ_0 cancels in expectation by the martingale property, leaving only a second-order contribution of order $\|\sigma_0\|_F^2$. This explicit isolation of the common-noise contribution is the technical core of the paper and, to our knowledge, has not been made explicit elsewhere in the literature.

The explicit constant has two immediate downstream uses, which we discuss in Section 7: it determines a sufficient discretisation step size for numerical schemes approximating the equilibrium, and it determines a sufficient learning-rate separation for multi-timescale reinforcement learning algorithms that learn the equilibrium from simulated or observed trajectories. We do not pursue these applications in technical detail here; they are treated in a companion paper on the numerical scheme and a companion paper on the learning algorithm (both in preparation).

Related work. Beyond Carmona and Delarue (2018) and Ahuja et al. (2019), quantitative stability for common-noise MFGs has been studied under additional regularity by Cardaliaguet et al. (2025), who establish regularity results under degenerate idiosyncratic noise. Our contribution is complementary: rather than regularity of the value function, we isolate the explicit dependence of the contraction rate on the model’s Lipschitz and common-noise constants, which is the quantity of direct use for algorithm design.

Limitations, stated upfront. The contraction established here is at the level of *expected* Wasserstein distance, not pathwise distance (Remark 4.2); it relies on stability of optimal transport maps, which in turn requires absolute continuity of the measures involved (Assumption 2.2); and the constant C_{reg} is fully explicit only in specific tractable models (e.g., the linear-quadratic case), remaining an implicit bound on Lions-derivative regularity in the general nonlinear case. We return to these points in Section 7.

Organisation. Section 2 states the model and standing assumptions. Section 3 defines the mean-field fixed-point map. Section 4 states the main theorem. Section 5 proves the stability estimate via synchronous coupling, with the most technical algebraic steps deferred to Appendix A. Section 6 concludes existence, uniqueness, and the ε -Nash property. Section 7 discusses the contraction constant and the result’s limitations.

2 Model and Standing Assumptions

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ supporting a d -dimensional idiosyncratic Brownian motion W^i and an independent d_0 -dimensional common-noise Brownian motion W^0 . A representative agent's state $X_t \in \mathbb{R}^d$ evolves according to the McKean–Vlasov SDE

$$dX_t = b(t, X_t, \mu_t, \alpha_t) dt + \sigma dW_t^i + \sigma_0 dW_t^0, \quad X_0 \sim \mu_{\text{init}}, \quad (1)$$

where $\alpha_t \in A \subset \mathbb{R}^k$ is the agent's control, $\mu_t = \mathcal{L}(X_t \mid \mathcal{F}_t^{W^0})$ is the conditional law of the state given the common noise, $\sigma > 0$ is constant idiosyncratic volatility, and $\sigma_0 \in \mathbb{R}^{d \times d_0}$ is the common-noise coefficient. The agent maximises

$$J^\mu(\alpha) = \mathbb{E} \left[\int_0^T f(t, X_t, \mu_t, \alpha_t) dt + g(X_T, \mu_T) \right]. \quad (2)$$

We write $\mathcal{M} := D([0, T]; \mathcal{P}_2(\mathbb{R}^d))$ for the Skorokhod space of càdlàg measure-valued processes adapted to $(\mathcal{F}_t^{W^0})$, equipped, for $\lambda > 0$, with the weighted norm

$$\|\mu^1 - \mu^2\|_{\lambda, T} := \sup_{t \leq T} e^{-\lambda t} \mathbb{E} [W_2(\mu_t^1, \mu_t^2)^2]^{1/2}. \quad (3)$$

Assumption 2.1 (Lipschitz drift). *There exists $L_b > 0$ such that for all $t, x, x', \mu, \mu', \alpha, \alpha'$,*

$$\|b(t, x, \mu, \alpha) - b(t, x', \mu', \alpha')\| \leq L_b (\|x - x'\| + W_2(\mu, \mu') + \|\alpha - \alpha'\|).$$

The running and terminal rewards f, g are Lipschitz with constants L_f, L_g .

Assumption 2.2 (Regularity). *b, f, g are C^1 in x and Lions-differentiable in μ (in the sense of Carmona and Delarue, 2018, Ch. 5); their Lions derivatives are bounded and Lipschitz in (x, μ) , with bounded second-order Lions derivatives where required. The initial law $\mu_{\text{init}} \in \mathcal{P}_2(\mathbb{R}^d)$ has finite fourth moment.*

Assumption 2.3 (Bounded common noise; uniform concavity). $\|\sigma_0\|_F \leq M_\sigma < \infty$. *The map $\alpha \mapsto H(t, x, \mu, p, \alpha) := p \cdot b(t, x, \mu, \alpha) + f(t, x, \mu, \alpha)$ is uniformly concave with modulus $\lambda_H > 0$.*

These assumptions are standing throughout the paper and correspond to the structural conditions under which Ahuja et al. (2019) establish qualitative well-posedness; our contribution is the explicit quantitative refinement under the same hypotheses.

3 The Mean-Field Fixed-Point Map

For fixed $\mu \in \mathcal{M}$, the representative agent's problem is a standard stochastic control problem. By the stochastic maximum principle, the unique optimal control α^μ is characterised by the forward-backward system

$$\begin{aligned} dX_t^\mu &= b(t, X_t^\mu, \mu_t, \alpha_t^\mu) dt + \sigma dW_t^i + \sigma_0 dW_t^0, & X_0^\mu &\sim \mu_{\text{init}}, \\ -dp_t^\mu &= \partial_x H(t, X_t^\mu, \mu_t, p_t^\mu, \alpha_t^\mu) dt - q_t^\mu dW_t^i - r_t^\mu dW_t^0, & p_T^\mu &= \partial_x g(X_T^\mu, \mu_T), \\ \alpha_t^\mu &= \arg \max_{\alpha \in A} H(t, X_t^\mu, \mu_t, p_t^\mu, \alpha). \end{aligned} \quad (4)$$

Existence and uniqueness of (4) for each fixed μ follows from standard results for McKean–Vlasov FBSDEs with Lipschitz coefficients (Carmona and Delarue, 2018, Ch. 4), or via the monotone-functional continuation method of Ahuja et al. (2019). We define the *mean-field map*

$$\Phi(\mu)_t := \mathcal{L}(X_t^\mu \mid \mathcal{F}_t^{W^0}), \quad t \in [0, T], \quad (5)$$

so that $\Phi : \mathcal{M} \rightarrow \mathcal{M}$, and an MFG equilibrium is precisely a fixed point $\mu^* = \Phi(\mu^*)$.

4 Main Theorem

Theorem 4.1 (Quantitative well-posedness). *Under Assumption 2.1–2.3, there exists a unique solution to the FBSDE system (4) with $\mu = \Phi(\mu)$. Moreover, for any $\mu^1, \mu^2 \in \mathcal{M}$,*

$$\mathbb{E}[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2] \leq C_0 \int_0^t e^{C(t-s)} \mathbb{E}[W_2(\mu_s^1, \mu_s^2)^2] ds, \quad (6)$$

with

$$C = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}, \quad C_0 = L_b + C_\alpha L_b, \quad (7)$$

where C_α is the Lipschitz constant of α^μ in (x, p, μ) and C_{reg} depends on the Lipschitz constants of the Lions derivatives of b, f, g and the linear BSDE estimate constant. Consequently, for any $\lambda > C/2$, Φ is a contraction on $(\mathcal{M}, \|\cdot\|_{\lambda, T})$ with factor

$$\rho = \sqrt{\frac{C_0}{2\lambda - C}} < 1, \quad (8)$$

and the unique fixed point $\mu^* = \Phi(\mu^*)$ is the MFG equilibrium. The ε -Nash property holds for the associated N -player game with rate $\varepsilon_N = O(N^{-1/2})$.

Remark 4.2 (Expectation vs. pathwise). The contraction (6) holds for the *expected* squared Wasserstein distance. The first-order stochastic term driven by σ_0 cancels in expectation by the martingale property under synchronous coupling (Section 5), but does not cancel pathwise; we do not claim, and the proof does not give, a pathwise contraction.

5 Proof of the Stability Estimate

5.1 Synchronous coupling

Fix $\mu^1, \mu^2 \in \mathcal{M}$ and let $(X^1, p^1, q^1, r^1, \alpha^1)$, $(X^2, p^2, q^2, r^2, \alpha^2)$ solve (4) with μ^1, μ^2 respectively. We couple the two systems on the same probability space, driving both with the *same* realisations of W^i and W^0 , and with optimally coupled initial conditions $X_0^1 = X_0^2$ (since both share the law μ_{init}). Write $\Delta X_t := X_t^1 - X_t^2$, $\Delta p_t := p_t^1 - p_t^2$. By (1) and Assumption 2.1,

$$d(\Delta X_t) = [b(t, X_t^1, \mu_t^1, \alpha_t^1) - b(t, X_t^2, \mu_t^2, \alpha_t^2)] dt. \quad (9)$$

Crucially, the diffusion coefficients σ, σ_0 are state- and measure-independent, so they cancel *exactly* (not merely in expectation) in ΔX_t : the synchronous coupling removes the noise term from the difference process entirely at this stage. The common noise instead enters the estimate through its effect on the *adjoint* difference Δp_t , via the BSDE (4), as we make precise below.

5.2 Itô expansion and Gronwall setup

By Itô's formula applied to $|\Delta X_t|^2$ and the Lipschitz bound (Assumption 2.1) on the drift difference,

$$\frac{d}{dt} \mathbb{E}[|\Delta X_t|^2] \leq (2L_b + L_b^2) \mathbb{E}[|\Delta X_t|^2] + L_b \mathbb{E}[W_2(\mu_t^1, \mu_t^2)^2] + L_b C_\alpha^2 (\mathbb{E}[|\Delta X_t|^2] + \mathbb{E}[|\Delta p_t|^2] + \mathbb{E}[W_2(\mu_t^1, \mu_t^2)^2]), \quad (10)$$

where the control-difference Lipschitz bound $\|\Delta \alpha_t\| \leq C_\alpha (\|\Delta X_t\| + \|\Delta p_t\| + W_2(\mu_t^1, \mu_t^2))$ follows from the implicit function theorem applied to the optimality condition $\partial_\alpha H = 0$ under the uniform concavity of Assumption 2.3 (full derivation in Appendix A.1).

5.3 Adjoint difference and the role of σ_0

The adjoint difference Δp_t satisfies a linear BSDE driven by the differences in H 's derivatives. The standard linear BSDE energy estimate (El Karoui et al., 1997, Prop. 2.1) gives

$$\mathbb{E} \left[\sup_t |\Delta p_t|^2 \right] \leq C_{\text{BSDE}} \mathbb{E} \left[|\Delta X_T|^2 + \int_0^T (|\Delta X_t|^2 + |\Delta \alpha_t|^2 + W_2(\mu_t^1, \mu_t^2)^2) dt \right], \quad (11)$$

with $C_{\text{BSDE}} = C'_0(1 + \|\sigma_0\|_F^2)$ for a constant C'_0 depending only on T and the Lipschitz constants of H 's second derivatives. The dependence on $\|\sigma_0\|_F^2$ arises because the common-noise martingale term $\int \Delta r_t dW_t^0$ in the BSDE energy identity contributes a quadratic variation term scaled by $\|\sigma_0\|_F^2$ through the martingale representation theorem (full derivation in Appendix A.2). This is the precise mechanism by which the common-noise coefficient enters the contraction constant: *not* through a first-order term in the state difference (which cancels exactly under synchronous coupling, as noted above), but through the BSDE estimate governing the adjoint process.

5.4 Combining the estimates

Substituting (11) into (10) and applying Gronwall's inequality to $y_t := \mathbb{E}[|\Delta X_t|^2] + \mathbb{E}[|\Delta p_t|^2]$ yields the single differential inequality

$$\frac{d}{dt} y_t \leq C_1 y_t + C_2 \mathbb{E}[W_2(\mu_t^1, \mu_t^2)^2],$$

with $C_1 = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}$ and $C_2 = L_b + C_\alpha L_b + C_{\text{BSDE}} L_\mu$, where C_{reg} collects the remaining Lipschitz/regularity terms not already isolated in the leading $2L_b^2 + 2\|\sigma_0\|_F^2$. Finally, since $\Phi(\mu^j)_t = \mathcal{L}(X_t^j \mid \mathcal{F}_t^{W^0})$, $W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2 \leq \mathbb{E}[|\Delta X_t|^2 \mid \mathcal{F}_t^{W^0}]$, and taking expectations and applying Gronwall's lemma to y_t gives (6) with $C = C_1$, $C_0 = C_2$, after absorbing the boundary term via $y_0 = 0$. This proves Theorem 4.1's stability estimate; full algebraic detail is given in Appendix A.

6 Existence, Uniqueness, and the ε -Nash Property

Since $(\mathcal{M}, \|\cdot\|_{\lambda, T})$ is complete (Billingsley, 1999), the contraction (8) and the Banach fixed-point theorem give a unique fixed point $\mu^* \in \mathcal{M}$, establishing existence and uniqueness of the FBSDE system's solution.

For the ε -Nash property, let $\mu_t^{N, -i}$ be the empirical measure of $N - 1$ agents following the equilibrium strategy α^* , while agent i deviates. By the conditional propagation of chaos (Carmona and Delarue, 2018, Thm 6.16), $\mathbb{E}[W_2(\mu_t^{N, -i}, \mu_t^*)^2] \leq C_{\text{PoC}}/N$ for a constant independent of N . Combining this with the Lipschitz stability of the rewards and (6) applied to the finite- N empirical measure gives

$$\sup_{\alpha^i} (J_i(\alpha^i, \alpha^{-i, *}) - J_i(\alpha^*, \alpha^{-i, *})) \leq \varepsilon_N = O(N^{-1/2}),$$

so the symmetric profile $(\alpha^*, \dots, \alpha^*)$ is an ε_N -Nash equilibrium of the N -player game.

7 Discussion

7.1 The contraction constant

Table 1 summarises the origin of each term in C .

Table 1: Components of the contraction constant C

Term	Expression	Origin
Drift Lipschitz	$2L_b^2$	Gronwall step on ΔX_t
Common noise	$2\ \sigma_0\ _F^2$	BSDE estimate constant C_{BSDE}
Regularity	C_{reg}	Lions-derivative and BSDE bounds

In the linear-quadratic case (drift linear in $(x, \bar{\mu}, \alpha)$, quadratic rewards), all constants – including C_{reg} – are explicit functions of the model parameters via the associated Riccati system; we give this specialisation in Appendix A.3 as a worked example. For general nonlinear coefficients, C_{reg} remains an implicit bound on Lions-derivative Lipschitz constants and BSDE operator norms.

7.2 Practical use of the explicit constant

The constant C directly bounds the rate at which numerical and learning errors can be amplified through the fixed-point iteration. A numerical scheme approximating Φ to discretisation error δ inherits, via (6), an equilibrium approximation error of order $\delta/(1 - \rho)$ with ρ as in (8); similarly, a multi-timescale learning algorithm’s slowest permissible timescale ratio is governed by C . We develop these applications, with full numerical and reinforcement-learning detail, in companion work currently in preparation.

7.3 Limitations

- **Expectation, not pathwise.** Remark 4.2 applies throughout; pathwise stability would require a different (and to our knowledge open) argument, since the cancellation we exploit is intrinsically an expectation-level phenomenon.
- **Regularity of optimal transport maps.** The argument implicitly uses stability properties of optimal couplings, which require absolute continuity of the measures involved. Discrete empirical measures – the practically relevant case in, e.g., financial applications – violate this, and a regularisation (kernel smoothing, entropic optimal transport) is needed in practice; the resulting additional approximation error is not quantified here.
- **Implicit C_{reg} .** Outside tractable special cases, C_{reg} is a bound, not a closed-form expression, on Lions-derivative regularity constants that are themselves typically estimated rather than known exactly in applications.
- **Homogeneous common-noise exposure.** The cancellation argument assumes all agents share the same common-noise loading σ_0 . If agents have heterogeneous exposures, the first-order cancellation under synchronous coupling fails and the argument must be revisited.

8 Conclusion

We have given an explicit, parameter-dependent contraction estimate for the mean-field fixed-point map governing common-noise MFG equilibria, isolating the precise mechanism (a second-order BSDE effect, not a first-order state-difference effect) by which the common-noise coefficient enters the contraction rate. The result refines existing qualitative well-posedness theory (Ahuja et al., 2019; Carmona and Delarue, 2018) to a quantitative statement usable directly in algorithm design, at the

cost of an expectation-level (rather than pathwise) guarantee and an implicit regularity constant outside tractable special cases. We view the extension to pathwise stability, and to heterogeneous common-noise exposure, as the most pressing open directions.

A Supplementary Proofs

A.1 Control-difference Lipschitz constant

Differentiating the optimality condition $\partial_\alpha H(t, X_t, \mu_t, p_t, \alpha_t^*) = 0$ implicitly in (x, p, μ) and using the uniform concavity bound $\partial_{\alpha\alpha} H \leq -\lambda_H < 0$ (Assumption 2.3), the implicit function theorem gives $\nabla_{(x,p,\mu)} \alpha^* = -(\partial_{\alpha\alpha} H)^{-1} \nabla_{(x,p,\mu)} \partial_\alpha H$, so $C_\alpha = (L_b + L_\mu)/\lambda_H$, where L_μ bounds the Lions derivative of f .

A.2 The BSDE constant C_{BSDE}

The linear BSDE for Δp_t has driver coefficients bounded by the second derivatives of H (Assumption 2.2). The energy identity for linear BSDEs (El Karoui et al., 1997, Prop. 2.1) expresses $\mathbb{E}[\sup_t |\Delta p_t|^2]$ in terms of the terminal condition and the running driver, plus the quadratic variation of the martingale terms $\int \Delta q_t dW_t^i + \int \Delta r_t dW_t^0$. The idiosyncratic term contributes a constant independent of σ_0 ; the common-noise term contributes $\|\sigma_0\|_F^2 \cdot (\text{driver bound})$ via the martingale representation theorem applied to the common-noise filtration. Collecting these gives $C_{\text{BSDE}} = C'_0(1 + \|\sigma_0\|_F^2)$.

A.3 Linear-quadratic specialisation

For $b(t, x, \mu, \alpha) = \kappa(\bar{\mu} - x) + \alpha$, $f = -(Qx^2 + \frac{\lambda}{2}\alpha^2)$, $g = -P(x - \bar{\mu}_T)^2$, the value function is quadratic, $u(t, x, \mu) = -\frac{1}{2}A_t x^2 - B_t x \bar{\mu} - \dots$, with (A_t, B_t) solving a scalar Riccati system. All constants in Theorem 4.1, including C_{reg} , are then explicit functions of $\kappa, \lambda, Q, P, \sigma, \sigma_0$ and the Riccati solution; we omit the lengthy but routine computation.

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